Wi-Fi is invisible. Coverage gaps are not.

Affordable spatial surveys for IT offices, laboratories and institutions.



1 Hidden weak zones

Signal bars cannot explain room-to-room variation.



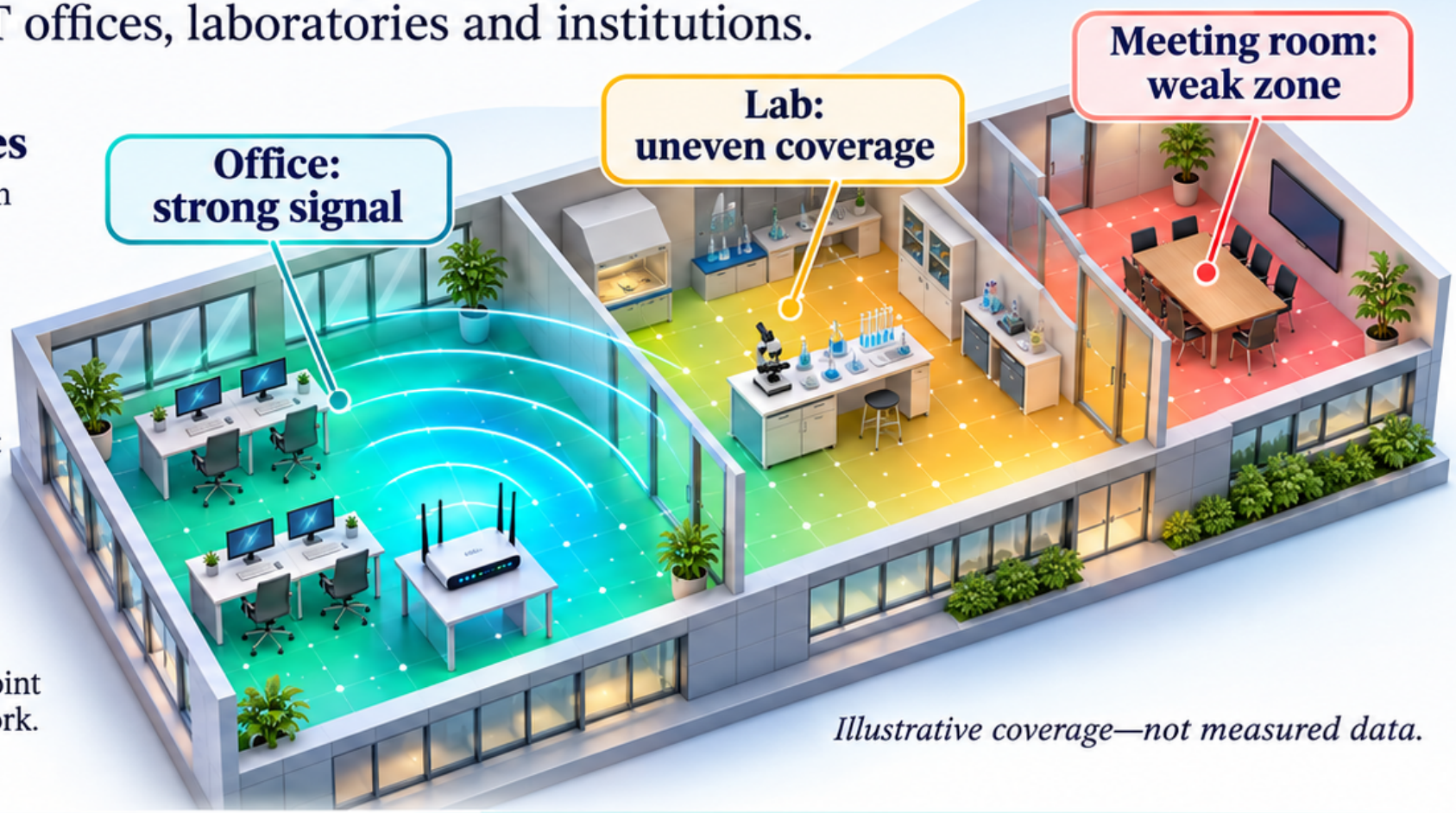
2 Costly survey workflows

Professional tools, specialist skills and repeat visits add cost.



3 Poor placement decisions

Without a map, access-point changes become guesswork.

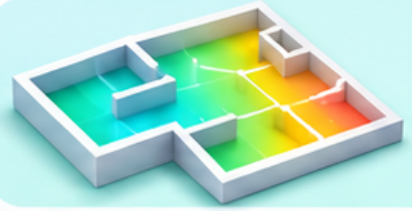


Illustrative coverage—not measured data.



Commercial example: ₹83,300 / year

Hamina Planner: US$980/year at illustrative ₹85/US$.



Our opportunity

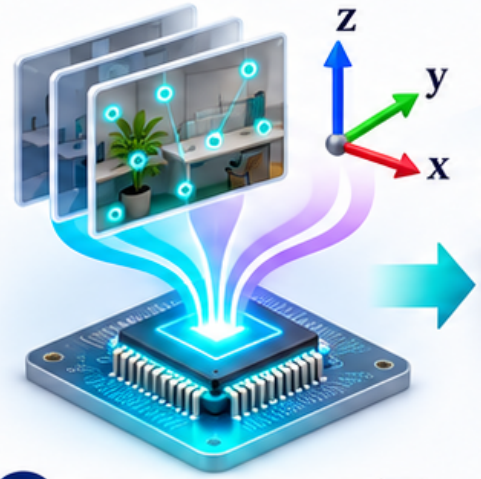
A portable, lower-cost way to see where Wi-Fi is strong or weak.

Vendor examples: Ekahau and Hamina—not an exhaustive list. Source: hamina.com/pricing, 4 Nov 2026. Taxes excluded; INR conversion is illustrative.

Walk the building. See Wi-Fi in 3D.

A portable Raspberry Pi 5 survey system with an interactive laptop view.







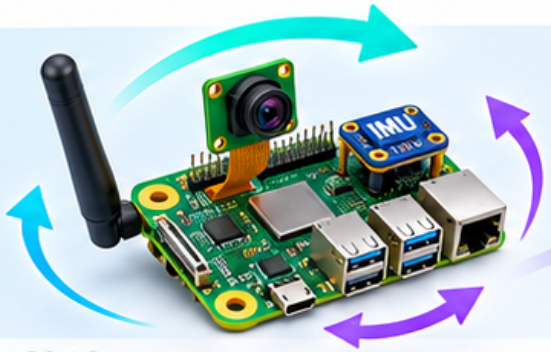


1 Capture
Wi-Fi RSSI + camera frames + IMU

2 Process on Pi 5
Estimate movement and orientation

3 Build the spatial map
Associate signal samples with positions

4 Explore on the laptop
Inspect coverage • Filter networks • Export



Move the device. Rotate the view.

Current: heading rotation. Planned: full pitch-and-roll tilt.

The IMU controls viewing orientation; the camera estimates movement.



Local edge processing | **Room-aware surveys** | **Accessible hardware**

Concept illustration • Full tilt and calibrated positioning are planned; accuracy and final hardware cost require validation.

From sensor measurements to spatial Wi-Fi insight.

Raspberry Pi 5 edge processing • HTTP over local network • Three.js browser visualization

RASPBERRY PI 5

RF — **Wi-Fi scans**
iw / iwlist • RSSI + AP identity

AP_01 −48 dBm
AP_02 −61 dBm
AP_03 −67 dBm

Acquire raw dBm

Camera — **Camera frames**
USB / V4L2

Visual odometry
ORB → LK tracking → RANSAC recoverPose → relative *R*, *t*

Spatial RF samples
(x, y, z, time, dBm)
Monocular scale is uncalibrated.

Interpolate coverage
IDW • Gaussian RBF • kernel methods

IMU — **6-axis IMU**
Accel + gyro • I²C

Orientation filter
Gyro bias + complementary filter

Heading API
80 ms polling • smoothed yaw

View rotation only

Flask / JSON

Three.js / WebGL
Rooms
RF heatmap
Walk path
SSID filters

Rooms • RF heatmap • Walk path
SSID filters • CSV / PNG

EXPERIMENTAL ROOM RECONSTRUCTION

4 room views → MiDaS depth + YOLOv8n → Pinhole + voxel reduction → RANSAC / SVD planes → Geometry JSON → /scene

ONNX Runtime • Approximate geometry; registration needs validation

Normalize: $I = clip((r + 90)/60, 0, 1)$ | **Display EMA:** $s(t) = 0.3\,I(t) + 0.7\,s(t-1)$ | **IDW:** $\hat{r}(p) = \Sigma w_i r_i / \Sigma w_i;\ w_i = 1/d_i^2$

PLANNED | Full pitch/roll tilt • Calibrated visual–inertial EKF / SLAM • Loop closure • Multi-floor height sampling

Concept hardware illustrations. Current architecture follows project code; planned methods are not yet integrated.